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Case Report

A Delayed Presentation of Bilateral Adrenal Hemorrhage Secondary to COVID-19

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ABSTRACT

Background/Objective: Bilateral adrenal hemorrhage is a rare cause of adrenal insufficiency. Cases have been reported of acute adrenal crisis with bilateral adrenal hemorrhage during acute coronavirus disease of 2019 (COVID-19). Our objective was to report a delayed presentation of acute adrenal crisis with bilateral adrenal hemorrhage 2 months after COVID-19.
Case Report: An 89-year-old man who was hospitalized for COVID-19 pneumonia 2 months prior presented with lethargy. He was disorientated and hypotensive to 70/50 mm Hg without improvement with intravenous fluids. According to his family, since his previous hospitalization for COVID-19, his mental status had continued to deteriorate, and he was no longer able to perform activities of daily living. A computed tomography scan of the abdomen revealed bilateral heterogeneous enlargement of the adrenal glands. Laboratory values were significant for an AM cortisol level of 8.42 mcg/dL, a sodium level of 134 mEq/L, and a bicarbonate level of 17 mEq/L. He was treated intravenously with hydrocortisone 100 mg and showed rapid improvement.
Discussion: It has been shown that COVID-19 disease may cause an increased risk of bleeding or thromboembolism. The exact frequency of bilateral adrenal hemorrhage secondary to COVID-19 is unknown. Although there are a handful of cases reported, there are none to our knowledge with a delayed presentation, as exhibited in our patient.
Conclusion: The patient's presentation was consistent with acute adrenal crisis due to bilateral adrenal hemorrhage from prior COVID-19 disease. We aimed to highlight the importance of clinicians being aware of adrenal hemorrhage and adrenal insufficiency as a possible delayed consequence in patients with a history of COVID-19.
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Introduction

The novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) caused the coronavirus disease of 2019 (COVID-19) global pandemic. Although initially shown to cause pulmonary disease, it quickly evolved into a myriad of extrapulmonary signs and symptoms. One of these conditions is primary adrenal insufficiency, as depicted by Hashim et al,[1] although its prevalence is yet to be established.[2] Bilateral adrenal hemorrhage is a rare cause of primary adrenal insufficiency that can be triggered by infections. Bilateral adrenal hemorrhage has been reported in association with COVID-19 in only a handful of case reports since 2020.[3] Even among the few case reports of COVID-19–associated adrenal insufficiency, this case is unusual because it demonstrates a delayed presentation of acute adrenal crisis 2 months after COVID-19.

Case Report

An 89-year-old man presented to the hospital for anorexia, lethargy, and confusion. He had a past medical history of prostate

Abbreviations: ACTH, adrenocorticotropic hormone; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; COVID-19, Coronavirus disease of 2019; IV, Intravenous.

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cancer status post brachytherapy, dementia, glaucoma, nephrolithiasis, and COVID-19 pneumonia. Two months before this presentation, the patient was admitted to the hospital after a mechanical fall at home that did not result in loss of consciousness or any major injuries. During this initial admission, the patient was incidentally diagnosed with COVID-19 pneumonia. The use of supplemental oxygen or antiviral medications was not required. However, he was treated for superimposed bacterial pneumonia with ceftriaxone and azithromycin owing to a productive cough. The patient's cough improved, and he was discharged home within 1 week. After this hospitalization, the patient's daughter noted that her father never returned to his previous level of well-being. He previously lived alone and was able to cook and clean for himself; however, he was now unable to perform his usual activities of daily living. He grew progressively weaker and often appeared confused beyond his baseline, such as forgetting the names of his family members, and was unable to go grocery shopping on his own. It was also reported that he had low blood pressure readings on ambulatory monitoring performed by his daughter, with systolic blood pressures between 70 and 90 mm Hg.

Two months after the aforementioned initial hospitalization during which COVID-19 was diagnosed, his family again sought care at the hospital owing to his progressive symptoms. Upon admission, his temperature was 38.0 °C, blood pressure was 100/66 mm Hg, pulse rate was 83 beats/min, respirations were 17/min, and saturation level was 96% on room air. He appeared dehydrated and was disoriented. Abdominal examination was significant for a right-sided reducible inguinal hernia and was only orientated to himself. The remainder of the physical examination were within normal limits. The results of an infectious workup were negative, including a normal white blood cell count, unremarkable chest x-ray, negative urinalysis, and eventually negative blood cultures. He was empirically treated with broad-spectrum antibiotics and with rapid administration of crystalloid solution (normal saline) to equal 30 mL/kg; however, he remained hypotensive, with blood pressures ranging from 70/50 mm Hg to 95/62 mm Hg. A computed tomography scan of the abdomen and pelvis with contrast was obtained to evaluate the inguinal hernia seen on physical examination, which revealed bilateral adrenal hemorrhages (Fig.) and a right inguinal hernia containing a small segment of large bowel without evidence of strangulation or obstruction. This raised concern for acute adrenal crisis. Serum cortisol levels were measured first, and then the patient was urgently treated with an additional 3-L normal saline bolus and given a dose of intravenous (IV) hydrocortisone 100 mg. Laboratory data were as follows: an AM cortisol level of 8.42 mcg/dL (range, 4.3-22 mcg/dL), a sodium level of 134 mEq/L (range, 135-146 mEq/L), and a bicarbonate level of 17 mEq/L (range, 21-32 mEq/L). Serum adrenocorticotropic hormone (ACTH) measurement was pending. After treatment with IV steroids, the patient's systolic blood pressure improved to 100 to 110 mm Hg. The patient continued receiving IV hydrocortisone 100 mg every 8 hours, with rapid improvement in his clinical status, becoming hemodynamically stable, and with a noted improvement in mental status. Hydrocortisone was then tapered and switched to oral maintenance dosing. Additionally, he was tested for gonorrhea, cytomegalovirus, Epstein-Barr virus, human immunodeficiency virus, and tuberculosis, and the results for all of these were negative for acute or previous infection. This led to the diagnosis of a delayed presentation of acute adrenal crisis in the setting of bilateral adrenal hemorrhage due to a previous SARS-CoV-2 infection.



Fig. Computed Tomography of the abdomen and pelvis with contrast revealing bilateral heterogeneous adrenal hemorrhages (arrows), measuring 7.0 cm × 4.5 cm on the right and 5.6 cm × 6 cm on the left.

Highlights

- Bilateral adrenal hemorrhage is a rare secondary outcome of COVID-19 pneumonia
- Bilateral adrenal hemorrhage can occur during acute illness or after resolution of COVID-19
- Acute adrenal insufficiency due to adrenal hemorrhage is diagnosed with imaging and biochemical studies
- Acute adrenal insufficiency should be treated promptly with fluids and hydrocortisone

Clinical Relevance

We present a case of bilateral adrenal hemorrhage that presented after an acute infection with severe acute respiratory syndrome coronavirus 2. Although there are a handful of case reports of adrenal hemorrhage during COVID-19, this is the first case report to our knowledge showcasing a delayed presentation of bilateral adrenal hemorrhage after COVID-19.

Discussion

Infectious etiologies of bilateral adrenal hemorrhage complicated by adrenal insufficiency are classically associated with Waterhouse-Friderichsen syndrome, which has been seen in bacterial septicemia due to *Streptococcus pneumoniae, Neisseria gonorrhoeae, Haemophilus influenzae, and Mycobacterium tuberculosum*, with many of the cases reported in children.[4,5] In adults, viral causes include HIV, Epstein-Barr virus, Parvovirus-B19, Cytomegalovirus, and, more recently, SARS-CoV-2.

Clinical features of acute adrenal insufficiency include hypotension, vomiting, abdominal pain, fatigue, and confusion, with metabolic derangements such as hyponatremia, hyperkalemia, and hypoglycemia.[6] Hyponatremia is a result of the loss of aldosterone-mediated sodium retention and hypersecretion of the antidiuretic hormone leading to water retention, and a reduction in the plasma sodium concentration.[7] As described in a review by Bornstein et al,[8] adrenal insufficiency in the setting of adrenal hemorrhage is suspected when the AM cortisol level is <5 μg/dL. In indeterminate cases (when the cortisol level is between 5 mcg/dL and 12 mcg/dL), dynamic testing via the cosyntropin stimulation test is indicated. An injection of 250 μg of cosyntropin (ACTH) is performed intravenously or intramuscularly, with cortisol levels measured 30

and 60 minutes after injection; failure to obtain a cortisol level of ≥18 μg/dL is suggestive of adrenal insufficiency. In the setting of possible acute adrenal crisis, it is recommended to initiate immediate therapy with glucocorticoid replacement, such as IV hydrocortisone, before diagnostic tests are resulted,[8] and maintenance dosing initiated thereafter, which was the scenario in our patient.

The etiology of bilateral adrenal hemorrhage is hypothesized to be due to the anatomical and physiologic characteristics of the adrenal glands. The adrenal glands have an abundant blood supply—3 main arteries, the superior, middle, and inferior suprarenal arteries, and a single adrenal vein. This vasculature is known as the "adrenal dam," which makes it susceptible to hemorrhage.[9,10] One postulation is that adrenal hemorrhage is a consequence of ACTH and catecholamines being secreted because of a stress response. This results in increased blood flow, vasoconstriction, and platelet aggregation, which leads to reperfusion and bleeding.[11,12] Bilateral adrenal hemorrhage in the setting of COVID-19 is thought to be due to a prothrombotic state. It has been shown that the SARS-CoV-2 virus enters the cells and binds angiotensin-converting enzyme receptors on endothelial cells. This results in a dysregulation of the coagulation associated with hypercoagulability, thus leading to venous and arterial thrombosis.[13]

In the literature, there have been several case reports of patients with acute adrenal crisis due to bilateral adrenal hemorrhage after COVID-19 disease; however, these individuals had underlying antiphospholipid antibodies.[14] This was demonstrated in a case series reported by Espinosa et al[14] that illustrated patients with antiphospholipid syndrome and symptoms of adrenal insufficiency. The results demonstrated decreased baseline cortisol levels in 98% of patients, increased ACTH hormone levels in 96% of patients, and a positive cosyntropin stimulation test result in 100% of patients, which confirmed their diagnosis of adrenal insufficiency. Additionally, Machado et al[3] reported a female patient with a known history of antiphospholipid syndrome who presented with acute adrenal crisis after COVID-19. However, our patient presented with a delayed appearance of adrenal insufficiency. He had a medical history significant for previous COVID-19 disease without a known underlying coagulopathy. This suggests that the SARS-CoV-2 virus may independently increase bleeding risk and result in hypercoagulability.

Significantly, the mortality rate of adrenal hemorrhage is approximately 15%.[9] It is evident that adrenal hemorrhage may have devastating consequences, including death, if not recognized in a timely manner. We present this case with the aim for clinicians to be aware of this potential complication of the novel coronavirus both during the acute illness and after its resolution.

Disclosure

The authors have no multiplicity of interest to disclose.

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